

HALBERD

A 12-week SysML v1 to v2 port,
run on the agile kit

Backlog, daily stand-ups, sprint burn-downs,
funding and earned value

Halberd is fictional and notional, and so is every number here. It is the kit's worked example: one solutions architect ports a legacy SysML v1 model to SysML v2 text under a DevSecOps pipeline, 5 October to 23 December 2026.

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1 The program

Halberd's legacy SysML v1 model holds 1,960 in-scope items across seven organizations, and 1,450 DOORS requirements. One solutions architect ports it to SysML v2 text, one segment at a time, starting Monday 5 October 2026: two-week sprints, a baseline tagged every four weeks, and a stand-up every working day. The status meeting hears the alphabet: the 26 metrics A–Z of docs/STATUS-METRICS.html. Thanksgiving and 24–25 December are holidays, so the port runs 56 working days.

Everything here is computed by the kit, not typed. The project is a real kit project, built by compiling one stand-up file per day through the kit's own ledger; rebuild it and this document with one command:

```
perl sim/halberd-gen.pl --pdf                # data/halberd, docs/HALBERD.html and .pdf
perl bin/ledger.pl -f data/halberd/scrumbal Sprint:3 # the ledger: 330 items done in sprint 3
perl agile.pl --today 2026-11-05 data/halberd    # the cockpit on a blocked day
```

2 Backlog

The tome is *Halberd SysML v2 port*; its epics are the seven segments, in the order they are ported. Each task ports one day's worth of v1 items, counted in items (the journal's unit). A carried task counts in the segment being worked when it was committed; the rows add up to all 1,960 items.

Epic	Segment	v1 items	Tasks	Worked
E	Systems Engineering & Integration	368	12	10-05 – 10-20
V	Interceptor Development	350	11	10-21 – 11-03
B	Battle Management & Fire Control	330	10	11-04 – 11-17
T	Test & Evaluation	292	7	11-18 – 11-25
G	Government Program Office	160	5	11-30 – 12-04
M	Manufacturing & Production	244	8	12-07 – 12-15
L	Logistics & Sustainment	216	6	12-16 – 12-23
Total		1,960	59	

Table 1: Segments as epics.

3 Release burn-down

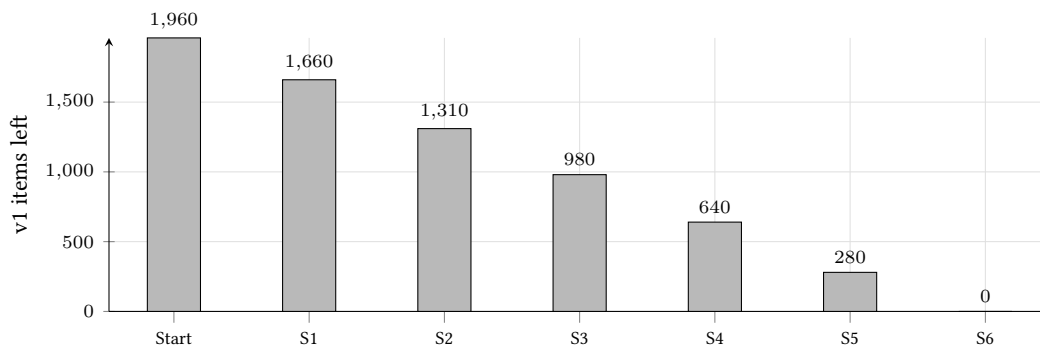


Figure 1: The whole v1 list, sprint by sprint.

Sprint	Dates	Committed	Done	Carried	Done % (X)	Items left
1	10-05 – 10-16	320	300	20	94	1,660
2	10-19 – 10-30	360	350	10	97	1,310
3	11-02 – 11-13	375	330	45	88	980
4	11-16 – 11-25	360	340	20	94	640
5	11-30 – 12-11	370	360	10	97	280
6	12-14 – 12-23	280	280	0	100	0

Table 2: Sprint totals, computed from the compiled journal by `Scrum::sprint_summary`; the generator stops if any differs from the doc.

4 Funding and earned value

A notional FY27 funding journal sits beside the stand-up journal, in the same ledger format (`data/halberd/funding.ledger`). Each segment is budgeted at a notional \$150 per v1 item, time-phased over the days the plan works it. \$323,400 is authorized on 1 October: the budget at completion of \$294,000 plus a 10% reserve. Labor posts weekly from the items actually ported; sprint 3 overran while it waited on the shared runner. Percent complete comes from the stand-ups, so earned value is computed, not estimated.

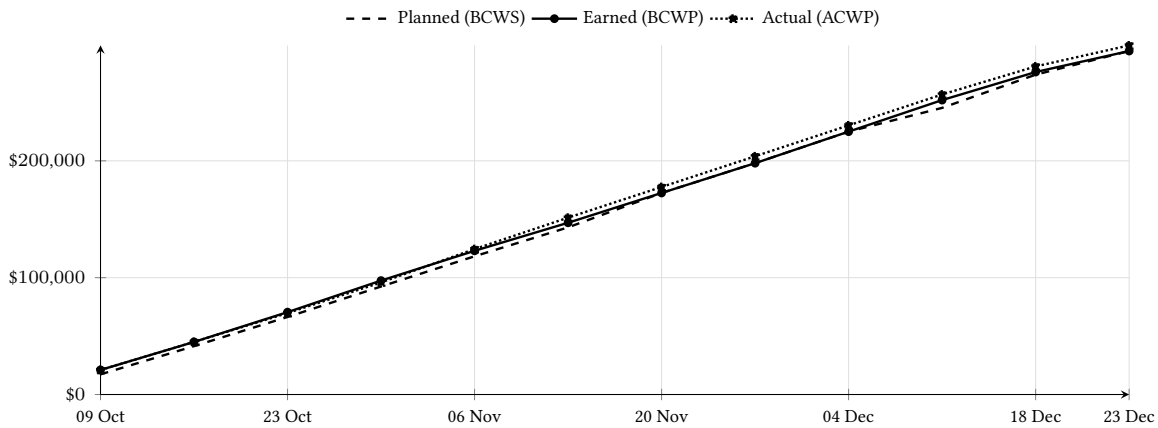


Figure 2: Cumulative cost at the end of each week.

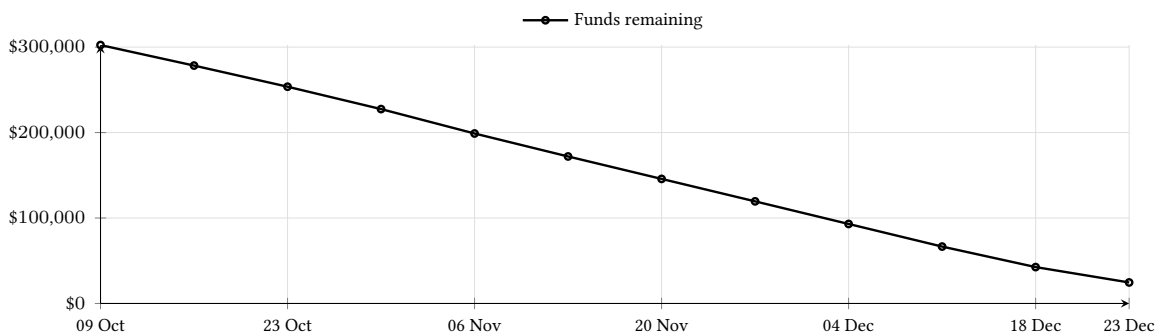


Figure 3: Funds remaining at the end of each week.

4.1 Earned value at 30 October 2026

Segment	Done	BAC	BCWS	BCWP	ACWP	CV	SV	CPI	SPI	EAC
E Systems Engineering & Integration	100%	\$55,200	\$55,200	\$55,200	\$54,894	\$306	\$0	1.01	1.00	\$54,894
V Interceptor Development	81%	\$52,500	\$37,234	\$42,300	\$41,031	\$1,269	\$5,066	1.03	1.14	\$50,925
B Battle Management & Fire Control	0%	\$49,500	\$0	\$0	\$0	\$0	\$0	–	–	–
T Test & Evaluation	0%	\$40,800	\$0	\$0	\$0	\$0	\$0	–	–	–
G Government Program Office	0%	\$27,000	\$0	\$0	\$0	\$0	\$0	–	–	–
M Manufacturing & Production	0%	\$36,600	\$0	\$0	\$0	\$0	\$0	–	–	–
L Logistics & Sustainment	0%	\$32,400	\$0	\$0	\$0	\$0	\$0	–	–	–
Total	33%	\$294,000	\$92,434	\$97,500	\$95,925	\$1,575	\$5,066	1.02	1.05	\$289,251

4.2 Earned value at 25 November 2026

Segment	Done	BAC	BCWS	BCWP	ACWP	CV	SV	CPI	SPI	EAC
E Systems Engineering & Integration	100%	\$55,200	\$55,200	\$55,200	\$54,894	\$306	\$0	1.01	1.00	\$54,894
V Interceptor Development	100%	\$52,500	\$52,500	\$52,500	\$52,455	\$45	\$0	1.00	1.00	\$52,455
B Battle Management & Fire Control	100%	\$49,500	\$49,500	\$49,500	\$54,522	–\$5,022	\$0	0.91	1.00	\$54,522
T Test & Evaluation	100%	\$40,800	\$40,800	\$40,800	\$42,024	–\$1,224	\$0	0.97	1.00	\$42,024
G Government Program Office	0%	\$27,000	\$0	\$0	\$0	\$0	\$0	–	–	–
M Manufacturing & Production	0%	\$36,600	\$0	\$0	\$0	\$0	\$0	–	–	–
L Logistics & Sustainment	0%	\$32,400	\$0	\$0	\$0	\$0	\$0	–	–	–
Total	67%	\$294,000	\$198,000	\$198,000	\$203,895	–\$5,895	\$0	0.97	1.00	\$302,753

Funds remaining \$119,505; average burn over the last two months \$95,925 a month; at that rate the money lasts 1.2 more months (to 2027-01-03).

4.3 Earned value at 23 December 2026

Segment	Done	BAC	BCWS	BCWP	ACWP	CV	SV	CPI	SPI	EAC
E Systems Engineering & Integration	100%	\$55,200	\$55,200	\$55,200	\$54,894	\$306	\$0	1.01	1.00	\$54,894
V Interceptor Development	100%	\$52,500	\$52,500	\$52,500	\$52,455	\$45	\$0	1.00	1.00	\$52,455
B Battle Management & Fire Control	100%	\$49,500	\$49,500	\$49,500	\$54,522	–\$5,022	\$0	0.91	1.00	\$54,522
T Test & Evaluation	100%	\$40,800	\$40,800	\$40,800	\$42,024	–\$1,224	\$0	0.97	1.00	\$42,024
G Government Program Office	100%	\$27,000	\$27,000	\$27,000	\$26,460	\$540	\$0	1.02	1.00	\$26,460
M Manufacturing & Production	100%	\$36,600	\$36,600	\$36,600	\$36,060	\$540	\$0	1.01	1.00	\$36,060
L Logistics & Sustainment	100%	\$32,400	\$32,400	\$32,400	\$32,400	\$0	\$0	1.00	1.00	\$32,400
Total	100%	\$294,000	\$294,000	\$294,000	\$298,815	–\$4,815	\$0	0.98	1.00	\$298,815

At the finish: \$298,815 spent against a \$294,000 budget (CPI 0.98); \$24,585 of the \$323,400 authorized is left, so the overrun came out of the reserve.

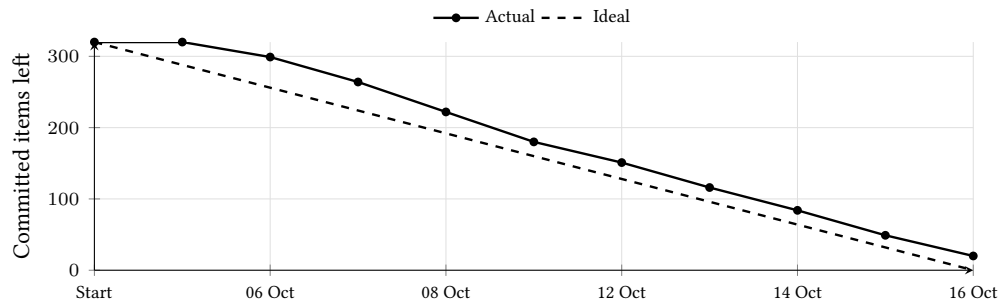
CPI = earned / actual (above 1 is under cost); SPI = earned / planned (above 1 is ahead); EAC = BAC / CPI. The same numbers come from bin/ledger.pl evm and forecast.

5 Sprints and daily stand-ups

Each stand-up answers yesterday, today and blockers, in the alphabet of the A–Z metrics. A dagger[†] marks a blocked day.

5.1 Sprint 1: 5 October 2026 to 16 October 2026

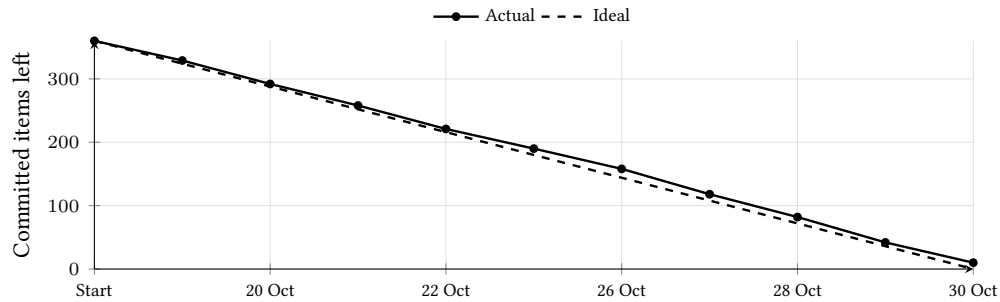
Committed 320. Done 300. Carried over 20. Done 94% (X).



Day	Ported	Left	Yesterday	Today	Blockers
Mon 05 Oct	0	1,960	Nothing yet; the pipeline goes up today.	Setting up the repo, the pre-commit gate and the CI lint job.	–
Tue 06 Oct	21	1,939	The pipeline is up: pre-commit gate and CI lint job, 0/0 to merge (E).	Continuing Systems Engineering & Integration.	–
Wed 07 Oct	35	1,904	21 more elements under configuration control, now 1% (A), all merged at 0/0 (E).	More of Systems Engineering & Integration goes under configuration control today.	–
Thu 08 Oct	42	1,862	35 more elements under configuration control, now 3% (A), all merged at 0/0 (E).	Bringing more of Systems Engineering & Integration under the pipeline.	–
Fri 09 Oct	42	1,820	42 more elements under configuration control, now 5% (A), all merged at 0/0 (E).	Continuing Systems Engineering & Integration.	–
Mon 12 Oct	29	1,791	42 more elements under configuration control, now 7% (A), all merged at 0/0 (E).	Importing the first DOORS module.	–
Tue 13 Oct	35	1,756	29 more elements under configuration control, now 9% (A), all merged at 0/0 (E). The first DOORS module is in, and requirements are now linking to model elements (C). 2 orphan and 4 unverified requirements, flagged to owners (F, G).	Bringing more of Systems Engineering & Integration under the pipeline.	–
Wed 14 Oct	32	1,724	35 more elements under configuration control, now 10% (A), all merged at 0/0 (E). 1 of 2 zone-crossing interfaces still need a security requirement (Q).	Continuing Systems Engineering & Integration.	–
Thu 15 Oct	35	1,689	32 more elements under configuration control, now 12% (A), all merged at 0/0 (E). 54 of 1,450 DOORS requirements linked (C).	More of Systems Engineering & Integration goes under configuration control today.	–
Fri 16 Oct	29	1,660	35 more elements under configuration control, now 14% (A), all merged at 0/0 (E). 3 interface mismatches caught at merge to date, 0 reached integration (H).	Bringing more of Systems Engineering & Integration under the pipeline.	–

5.2 Sprint 2: 19 October 2026 to 30 October 2026

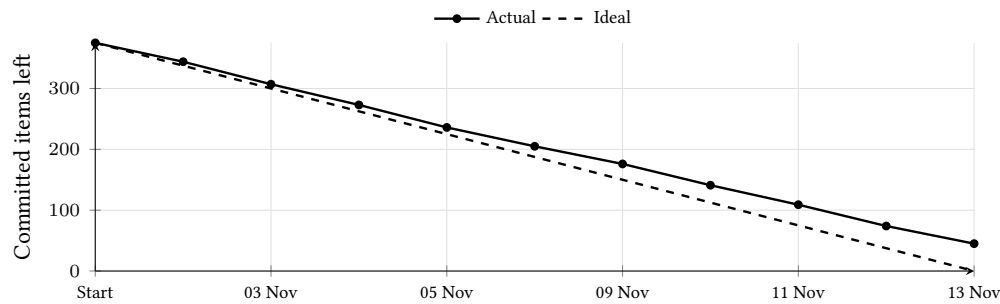
Committed 360. Done 350. Carried over 10. Done 97% (X). Includes HAL-10, 20 items carried in from sprint 1.



Day	Ported	Left	Yesterday	Today	Blockers
Mon 19 Oct	31	1,629	29 more elements under configuration control, now 15% (A), all merged at 0/0 (E). 12 orphan and 20 unverified requirements, flagged to owners (F, G).	Continuing Systems Engineering & Integration.	–
Tue 20 Oct	37	1,592	31 more elements under configuration control, now 17% (A), all merged at 0/0 (E). 2 of 5 zone-crossing interfaces still need a security requirement (Q).	Finishing Systems Engineering & Integration; it should be gated by end of day.	–
Wed 21 Oct	34	1,558	37 more elements under configuration control, now 19% (A), all merged at 0/0 (E). Systems Engineering & Integration is now gated on every merge: 1 of 7 segments (B). 138 of 1,450 DOORS requirements linked (C).	Bringing more of Interceptor Development under the pipeline.	–
Thu 22 Oct	37	1,521	34 more elements under configuration control, now 21% (A), all merged at 0/0 (E). 8 interface mismatches caught at merge to date, 0 reached integration (H).	Continuing Interceptor Development.	–
Fri 23 Oct	31	1,490	37 more elements under configuration control, now 22% (A), all merged at 0/0 (E). 36 orphan and 49 unverified requirements, flagged to owners (F, G).	More of Interceptor Development goes under configuration control today.	–
Mon 26 Oct	32	1,458	31 more elements under configuration control, now 24% (A), all merged at 0/0 (E). 3 of 7 zone-crossing interfaces still need a security requirement (Q).	Bringing more of Interceptor Development under the pipeline.	–
Tue 27 Oct	40	1,418	32 more elements under configuration control, now 26% (A), all merged at 0/0 (E). 233 of 1,450 DOORS requirements linked (C).	Continuing Interceptor Development.	–
Wed 28 Oct	36	1,382	40 more elements under configuration control, now 28% (A), all merged at 0/0 (E). 13 interface mismatches caught at merge to date, 0 reached integration (H).	More of Interceptor Development goes under configuration control today.	–
Thu 29 Oct	40	1,342	36 more elements under configuration control, now 29% (A), all merged at 0/0 (E). 56 orphan and 70 unverified requirements, flagged to owners (F, G).	Generating the interface table from the model.	–
Fri 30 Oct	32	1,310	40 more elements under configuration control, now 32% (A), all merged at 0/0 (E). The interface table is now generated from the model: 1 of 12 program documents (L), 0 hand edits (M). 2 of 9 zone-crossing interfaces still need a security requirement (Q).	Tagging baseline 1.	–

5.3 Sprint 3: 2 November 2026 to 13 November 2026

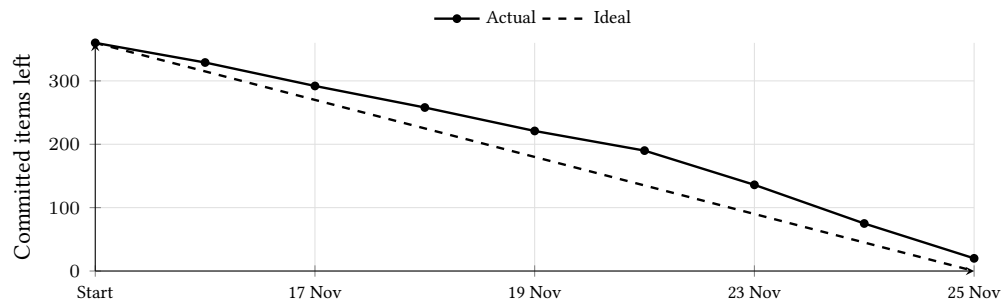
Committed 375. Done 330. Carried over 45. Done 88% (X). Includes HAL-21, 10 items carried in from sprint 2.



Day	Ported	Left	Yesterday	Today	Blockers
Mon 02 Nov	31	1,279	32 more elements under configuration control, now 33% (A), all merged at 0/0 (E). Baseline 1 is tagged: 650 elements (D), and it rebuilds identically (S). 340 of 1,450 DOORS requirements linked (C).	More of Interceptor Development goes under configuration control today.	–
Tue 03 Nov	37	1,242	31 more elements under configuration control, now 35% (A), all merged at 0/0 (E). 18 interface mismatches caught at merge to date, 0 reached integration (H).	Finishing Interceptor Development; it should be gated by end of day.	–
Wed 04 Nov [†]	34	1,208	37 more elements under configuration control, now 37% (A), all merged at 0/0 (E). Interceptor Development is now gated on every merge: 2 of 7 segments (B). 63 orphan and 78 unverified requirements, flagged to owners (F, G).	Continuing Battle Management & Fire Control.	Waiting on a shared runner. Gating locally with the pre-commit hook meanwhile (Y).
Thu 05 Nov [†]	37	1,171	34 more elements under configuration control, now 38% (A), all merged at 0/0 (E).	More of Battle Management & Fire Control goes under configuration control today.	Still waiting on the shared runner (Y).
Fri 06 Nov [†]	31	1,140	37 more elements under configuration control, now 40% (A), all merged at 0/0 (E). 454 of 1,450 DOORS requirements linked (C).	Bringing more of Battle Management & Fire Control under the pipeline.	Runner still pending; asked for it again (Y).
Mon 09 Nov	29	1,111	31 more elements under configuration control, now 42% (A), all merged at 0/0 (E). 25 interface mismatches caught at merge to date, 0 reached integration (H). The shared runner is live.	Continuing Battle Management & Fire Control.	–
Tue 10 Nov	35	1,076	29 more elements under configuration control, now 43% (A), all merged at 0/0 (E). 56 orphan and 72 unverified requirements, flagged to owners (F, G).	More of Battle Management & Fire Control goes under configuration control today.	–
Wed 11 Nov	32	1,044	35 more elements under configuration control, now 45% (A), all merged at 0/0 (E). 3 of 14 zone-crossing interfaces still need a security requirement (Q).	Generating the first ICD view and diffing it against the legacy ICD.	–
Thu 12 Nov	35	1,009	32 more elements under configuration control, now 47% (A), all merged at 0/0 (E). Drift checks found 47 discrepancies in the legacy ICDs, flagged for correction (T). 558 of 1,450 DOORS requirements linked (C).	Continuing Battle Management & Fire Control.	–
Fri 13 Nov	29	980	35 more elements under configuration control, now 49% (A), all merged at 0/0 (E). 30 interface mismatches caught at merge to date, 0 reached integration (H).	More of Battle Management & Fire Control goes under configuration control today.	–

5.4 Sprint 4: 16 November 2026 to 25 November 2026

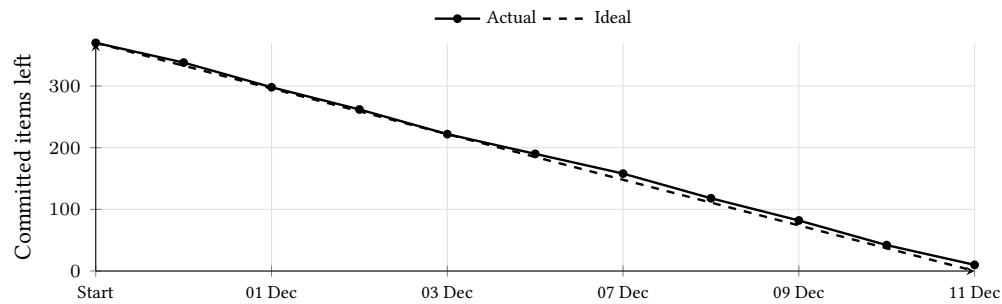
Committed 360. Done 340. Carried over 20. Done 94% (X). Includes HAL-32, 45 items carried in from sprint 3.



Day	Ported	Left	Yesterday	Today	Blockers
Mon 16 Nov	31	949	29 more elements under configuration control, now 50% (A), all merged at 0/0 (E). 47 orphan and 63 unverified requirements, flagged to owners (F, G).	Bringing more of Battle Management & Fire Control under the pipeline.	–
Tue 17 Nov	37	912	31 more elements under configuration control, now 52% (A), all merged at 0/0 (E). 1 of 16 zone-crossing interfaces still need a security requirement (Q).	Finishing Battle Management & Fire Control; it should be gated by end of day.	–
Wed 18 Nov	34	878	37 more elements under configuration control, now 53% (A), all merged at 0/0 (E). Battle Management & Fire Control is now gated on every merge: 3 of 7 segments (B). 3 of 7 segment owners now merge their own changes (Z). 670 of 1,450 DOORS requirements linked (C).	More of Test & Evaluation goes under configuration control today.	–
Thu 19 Nov	37	841	34 more elements under configuration control, now 55% (A), all merged at 0/0 (E). 36 interface mismatches caught at merge to date, 0 reached integration (H).	Bringing more of Test & Evaluation under the pipeline.	–
Fri 20 Nov	31	810	37 more elements under configuration control, now 57% (A), all merged at 0/0 (E). 51 orphan and 61 unverified requirements, flagged to owners (F, G).	Continuing Test & Evaluation.	–
Mon 23 Nov	54	756	31 more elements under configuration control, now 59% (A), all merged at 0/0 (E). 3 of 19 zone-crossing interfaces still need a security requirement (Q).	More of Test & Evaluation goes under configuration control today.	–
Tue 24 Nov	61	695	54 more elements under configuration control, now 61% (A), all merged at 0/0 (E). 804 of 1,450 DOORS requirements linked (C).	Wiring the T2 pre-flight prediction inputs to the model.	–
Wed 25 Nov	55	640	61 more elements under configuration control, now 65% (A), all merged at 0/0 (E). 64 of 64 pre-flight prediction parameters now come from the model (O), and 12 of 12 prediction runs name their baseline (P). 41 interface mismatches caught at merge to date, 0 reached integration (H).	Tagging baseline 2 before the holiday.	–

5.5 Sprint 5: 30 November 2026 to 11 December 2026

Committed 370. Done 360. Carried over 10. Done 97% (X). Includes HAL-40, 20 items carried in from sprint 4.

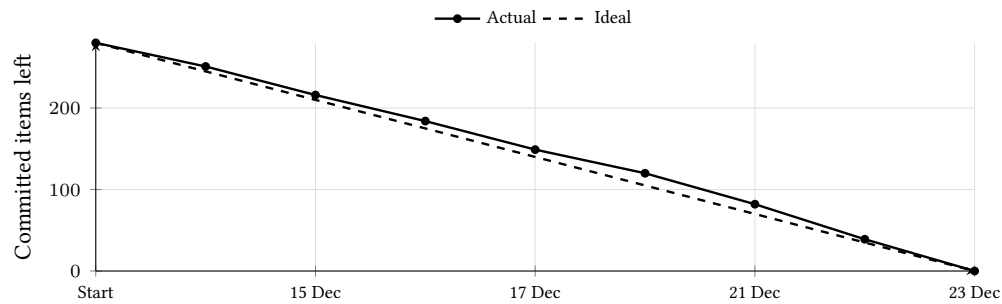


Day	Ported	Left	Yesterday	Today	Blockers
Mon 30 Nov	32	608	55 more elements under configuration control, now 67% (A), all merged at 0/0 (E). Test & Evaluation is now gated on every merge: 4 of 7 segments (B). 4 of 7 segment owners now merge their own changes (Z). Baseline 2 is tagged: 1,320 elements (D), rebuilds identically (S). 39 orphan and 48 unverified requirements, flagged to owners (F, G).	More of Government Program Office goes under configuration control today.	–
Tue 01 Dec	40	568	32 more elements under configuration control, now 69% (A), all merged at 0/0 (E).	Bringing more of Government Program Office under the pipeline.	–
Wed 02 Dec	36	532	40 more elements under configuration control, now 71% (A), all merged at 0/0 (E). 960 of 1,450 DOORS requirements linked (C).	Continuing Government Program Office.	–
Thu 03 Dec	40	492	36 more elements under configuration control, now 73% (A), all merged at 0/0 (E).	Reviewing threat and mitigation coverage with the ISSM.	–
Fri 04 Dec	32	460	40 more elements under configuration control, now 75% (A), all merged at 0/0 (E). 2 of 12 recorded threats lack a traced mitigation (R). 30 orphan and 37 unverified requirements, flagged to owners (F, G).	Measuring lead time on the last 10 merges.	–
Mon 07 Dec	32	428	32 more elements under configuration control, now 77% (A), all merged at 0/0 (E). Government Program Office is now gated on every merge: 5 of 7 segments (B). 5 of 7 segment owners now merge their own changes (Z). A model change now reaches a reviewed baseline in 4 days; the old change board cycle took 19 (N).	Continuing Manufacturing & Production.	–
Tue 08 Dec	40	388	32 more elements under configuration control, now 78% (A), all merged at 0/0 (E). 1,078 of 1,450 DOORS requirements linked (C).	More of Manufacturing & Production goes under configuration control today.	–
Wed 09 Dec	36	352	40 more elements under configuration control, now 80% (A), all merged at 0/0 (E). 47 interface mismatches caught at merge to date, 0 reached integration (H).	Generating the remaining ICD views.	–
Thu 10 Dec	40	312	36 more elements under configuration control, now 82% (A), all merged at 0/0 (E). 8 of 12 program documents now come straight from the model (L), about 30 labor hours saved per baseline (J). 21 orphan and 27 unverified requirements, flagged to owners (F, G).	Continuing Manufacturing & Production.	–

Day	Ported	Left	Yesterday	Today	Blockers
Fri 11 Dec	32	280	40 more elements under configuration control, now 84% (A), all merged at 0/0 (E). 0 of 25 zone-crossing interfaces still need a security requirement (Q).	More of Manufacturing & Production goes under configuration control today.	–

5.6 Sprint 6: 14 December 2026 to 23 December 2026

Committed 280. Done 280. Carried over 0. Done 100% (X). Includes HAL-51, 10 items carried in from sprint 5.



Day	Ported	Left	Yesterday	Today	Blockers
Mon 14 Dec	29	251	32 more elements under configuration control, now 86% (A), all merged at 0/0 (E). 1,210 of 1,450 DOORS requirements linked (C).	Bringing more of Manufacturing & Production under the pipeline.	–
Tue 15 Dec	35	216	29 more elements under configuration control, now 87% (A), all merged at 0/0 (E).	Finishing Manufacturing & Production; it should be gated by end of day.	–
Wed 16 Dec	32	184	35 more elements under configuration control, now 89% (A), all merged at 0/0 (E). Manufacturing & Production is now gated on every merge: 6 of 7 segments (B). 6 of 7 segment owners now merge their own changes (Z). 13 orphan and 18 unverified requirements, flagged to owners (F, G).	Rebuilding the gate after this morning's parser update.	–
Thu 17 Dec	35	149	32 more elements under configuration control, now 91% (A), all merged at 0/0 (E). The gate broke on a parser update and was restored in 2 hours (W). ICD views regenerate in 6 minutes (K).	Bringing more of Logistics & Sustainment under the pipeline.	–
Fri 18 Dec	29	120	35 more elements under configuration control, now 92% (A), all merged at 0/0 (E). 1,333 of 1,450 DOORS requirements linked (C).	Continuing Logistics & Sustainment.	–
Mon 21 Dec	38	82	29 more elements under configuration control, now 94% (A), all merged at 0/0 (E).	More of Logistics & Sustainment goes under configuration control today.	–
Tue 22 Dec	43	39	38 more elements under configuration control, now 96% (A), all merged at 0/0 (E). 6 orphan and 9 unverified requirements, flagged to owners (F, G).	Bringing more of Logistics & Sustainment under the pipeline.	–
Wed 23 Dec	39	0	43 more elements under configuration control, now 98% (A), all merged at 0/0 (E).	Tagging baseline 3.	–